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Ultra-broadband dual-polarization and arbitrary ratio power splitters based on Bezier curve optimized multimode interference

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We propose and demonstrate two types of 1×2 power splitters based on multimode interference (MMI), which are ultra-compact, fabrication friendly, and low loss. The contours of MMI and output tapers are optimized with Bezier curves, which can implement arbitrary ratio power splitters (ARPSs) and ultra-broadband dual-polarization power splitters (UDPSs). For ARPSs, the experimental results show that arbitrary power splitting ratios can be obtained with an average excess loss (*EL*) of 0.17 dB at 1550 nm for fundamental TE polarization. For UDPSs, the experimental results show that the *EL*s for fundamental TE and TM polarization are less than 0.63 dB and 0.44 dB over a large bandwidth of 415 nm (1260–1675 nm). The footprints of the proposed devices are less than $10 \mu\text{m} \times 2.5 \mu\text{m}$ (without input straight waveguide) with large fabrication tolerance. © 2023 Optica Publishing Group

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A power splitter, especially 1×2 splitter, is a fundamental element in photonic integrated circuits, which is desired to be compact, broadband, low loss, and fabrication friendly. The traditional 1×2 power splitters generally work in a specific wavelength band and polarization state with a 3-dB splitting ratio. However, in many applications, such as wavelength multiplexing/demultiplexing [1], optical switches [2], and optical phased arrays [3], arbitrary ratio power splitters (ARPSs) and ultra-broadband dual-polarization power splitters (UDPSs) are also required.

In recent years, various types of structures have been reported to obtain ARPSs, including directional couplers (DCs) [4–6], adiabatic tapers [7], and multimode interferometers (MMIs) [8–11]. DCs are widely used due to simplicity and low loss, but suffer from a small fabrication tolerance. The waveguide widths accuracy should be controlled within $\pm 10 \text{ nm}$ [5,6], which is difficult to obtain owing to the continuous and accurate adjustment of the splitting ratio. An adiabatic taper-based ARPS can consider both low loss and large fabrication tolerance but at the expense of footprint [7]. Different from the previous two types, MMIs can achieve arbitrary splitting ratios by introducing asymmetry with better tolerance to width variations

and compact footprint, but the excess loss (*EL*) will be much larger [9,10].

As for UDPSs, the adiabaticity directional couplers (ADCs) are a simple scheme to achieve broadband polarization-insensitive power splitter. However, they usually require long device lengths (usually $> 100 \mu\text{m}$ for TE mode) to avoid unexpected coupling and obtain uniform 3-dB power splitting [12]. To shorten the evolution lengths, researchers proposed quasi-adiabatic polarization-independent 3-dB couplers with the length of coupling region of $28.1 \mu\text{m}$ [12] and $11.7 \mu\text{m}$ [13]. However, the operation bandwidth is less than 200 nm. Subwavelength grating (SWG) structures have been employed in MMIs [14] and DCs [15] to achieve a compact footprint and large bandwidth characteristics for only one polarization. In Ref. [16], an anisotropic metamaterial based SWG-MMI is used for broadband power splitting. However, the typical feature size for such devices is $\sim 100 \text{ nm}$ [14,16], which needs high fabrication precision. Overall, it is still challenging to realize high-performance and fabrication-friendly ARPS and UDPS on the silicon-on-insulator (SOI) platform.

In this paper, we propose and demonstrate 1×2 power splitters with Bezier curve contours in the MMI section and output tapers on a 220-nm SOI platform. Arbitrary power splitting ratios could be achieved by adjusting the MMI length and the Bezier curve on one side of the MMI section. Experimental results indicate that the *EL*s are below 0.69 dB over a bandwidth from 1520 nm to 1590 nm for TE polarization. The demonstrated UDPSs could cover the whole optical communication band (1260–1675 nm) with the measured *EL*s below 0.63 dB for TE polarization and 0.44 dB for TM polarization. Both the ARPSs and UDPSs can maintain high performance with width variations of $\pm 40 \text{ nm}$, and such a large fabrication tolerance can be easily guaranteed by modern fabrication technology.

The schematic diagram of the proposed Bezier curve optimized MMI is shown in Fig. 1(a). The SOI platform with the 220-nm core silicon layer, 2- μm buried oxide layer, and 1- μm top oxide cladding is considered in this work. The contours of the MMI section and two output tapers are three types of cubic Bezier curves: Bezier curve A, Bezier curve B, and Bezier curve C, indicated by orange, green, and blue, respectively. Each cubic Bezier curve includes two end points and two curve points,

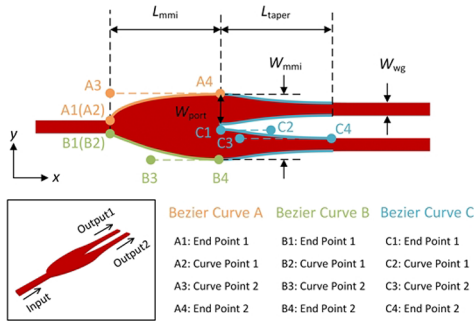


Fig. 1. Top view and 3D (inside the black box) schematic of the proposed 1×2 power splitter with Bezier curve contours in the MMI section and output tapers. (Three types of cubic Bezier curves are indicated by orange, green, and blue; the solid circle of the corresponding color represents the end points or curve points.)

which determine the shape of the curves. The shape control of the Bezier curve is convenient and intuitive. The shape can be simply adjusted by controlling the coordinates of the curve points with the fixed end points. Here, we make end point A1 and curve point A2 (end point B1 and curve point B2) coincide and locate on the upper (lower) edge of the input waveguide to ensure high-order modes excitation in the MMI section and reduce the optimization complexity. To ensure a smooth connection and reduce the unexpected reflections, the end point A4 (B4) and curve point A3 (B3) have the same y coordinates, as well as the end point C1 (C4) and curve point C2 (C3) have the same y coordinates.

The three-dimensional finite-difference time-domain (3D-FDTD) method is performed to simulate and optimize these devices. To obtain arbitrary power splitting ratios, it is necessary to separately adjust Bezier curve A and Bezier curve B with different shapes to introduce asymmetry in the MMI section. To simplify the optimization, we fix L_{A3-A4} to the same length as L_{mmi} and adjust L_{B3-B4} and L_{mmi} . The n is defined as the proportion of L_{B3-B4} in L_{mmi} , $n = L_{B3-B4} / L_{mmi}$. Here, we denote the power splitting ratios (PS) between the outputs as

$$PS = \frac{T_{Output1}}{T_{Output1} + T_{Output2}} \quad (1)$$

and the excess loss (EL) as

$$EL = -10 \log_{10}(T_{Output1} + T_{Output2}), \quad (2)$$

where $T_{Output1}$ and $T_{Output2}$ are the transmittances at the Output1 and Output2 ports, respectively. The widths of the MMI (W_{mmi}) and the input/output waveguide (W_{wg}) are chosen to be $2.5 \mu\text{m}$ and 500 nm , respectively. The minimum gap between the two output tapers is set to be 200 nm for easy fabrication. Figure 2(a) shows the calculated PS at 1550 nm with the L_{mmi} varying from $3.5 \mu\text{m}$ to $5.5 \mu\text{m}$ and n from 0 to 1. Different (n, L_{mmi}) correspond to different PS . A smaller n value means a greater enhancement of the asymmetry in the MMI section, and the more the PS deviates from 0.5. The calculated maximum EL with the wavelength from 1500 nm to 1600 nm is shown in Fig. 2(b). The black dotted line is the minimum EL with different n [also shown in Figs. 2(a) and 2(c)], covering the PS ranging from 0.5 to 0.9. A PS greater than 0.9 could be obtained with negative values of n . From Fig. 2(b), one can find that the simulated EL s could be below 0.3 dB when (n, L_{mmi}) is located on the

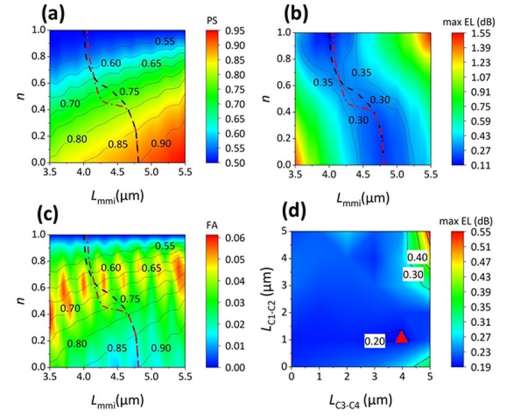


Fig. 2. (a) Calculated PS at 1550 nm ; (b) maximum EL in the working wavelength band of $1500\text{--}1600 \text{ nm}$; (c) FA with the L_{mmi} varying from $3.5 \mu\text{m}$ to $5.5 \mu\text{m}$ and n from 0 to 1; (d) calculated maximum EL in the wavelength band of $1500\text{--}1600 \text{ nm}$ for different output taper parameters. [Here, $(n, L_{mmi}) = (0.01, 4.81 \mu\text{m})$.]

black dotted line. In addition to EL , PS fluctuation amplitude (FA) within the working bandwidth is also a key parameter for ARPS. Hence, we calculate the FA with different (n, L_{mmi}) , as shown in Fig. 2(c). The FA represents the difference between the maximum PS and minimum PS within the working wavelength range ($1500\text{--}1600 \text{ nm}$ is considered in this work), i.e., $FA = \max(PS) - \min(PS)$. When n is approximately 0.6, the FA , located on the black dotted line, is larger than the surrounding parameters. Therefore, there is a trade-off between FA and EL . For balance, we slightly modify the black dotted line to avoid the large FA region and have negligible impact on EL , as shown by the red dotted line in Figs. 2(b) and 2(c). Thus, the FA is less than 0.03, and the EL is less than 0.35 dB within the wavelength range of $1500\text{--}1600 \text{ nm}$. We further optimize the output taper curves to achieve low EL . The taper length L_{taper} is set to be $5 \mu\text{m}$. The output tapers are symmetrical about the waveguide center. The device with (n, L_{mmi}) of $(0.01, 4.81 \mu\text{m})$, corresponding to PS of 0.9, is chosen to calculate the influence of L_{C1-C2} and L_{C3-C4} on the overall EL . The calculated results are shown in Fig. 2(d). The values in the figure correspond to the maximum EL in the working bandwidth of the corresponding parameters. The low loss of $<0.2 \text{ dB}$ can be obtained when $L_{C1-C2} = 1 \mu\text{m}$ and $L_{C3-C4} = 4 \mu\text{m}$ [red triangle in Fig. 2(d)].

In addition to ARPS, the proposed method can also be optimized for the implementation of UDPSSs. Unlike ARPS, 3-dB beam splitters need to maintain the symmetry of the device. Therefore, Bezier curve A and Bezier curve B need to maintain the same parameters. To simplify the optimization, we fix L_{A3-A4} (L_{B3-B4}) and L_{mmi} at the same length. The wavelength sensitivity of the MMI beat length is related to the square of the MMI width [17]. A narrow width is beneficial to achieve broad bandwidth with low excess loss. However, a wider width can obtain clearer imaging points reducing the reflection caused by the scattering on the gap [18]. Based on the above considerations, the scanning range of W_{mmi} is selected to be from $2.1 \mu\text{m}$ to $2.9 \mu\text{m}$. Then, the EL s for TE polarization and TM polarization with different L_{mmi} and W_{mmi} are also calculated by 3D-FDTD. In consideration of the two polarization states, FOM is selected as the average value of the maximum losses for two polarized lights over the wavelength band of $1260\text{--}1675 \text{ nm}$, i.e., $FOM = (\max(EL_{TE}) + \max(EL_{TM}))/2$. The calculated results are shown in Fig. 3(a).

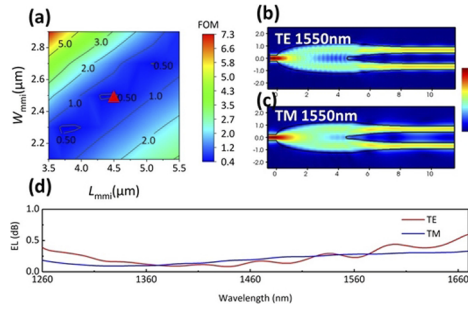


Fig. 3. (a) Calculated FOM with the L_{mmi} varying from 3.5 μm to 5.5 μm and W_{mmi} 2.1 μm to 2.9 μm ; propagation mode field for (b) TE polarization and (c) TM polarization at 1550 nm; (d) calculated EL s spectra of TE and TM polarization in the wavelength band of 1260–1675 nm.

The FOM is below 0.5 dB when the $(L_{\text{mmi}}, W_{\text{mmi}})$ is approximately (3.8 μm , 2.3 μm), (4.5 μm , 2.5 μm), and (5.1 μm , 2.7 μm). We then choose $L_{\text{mmi}} = 4.5 \mu\text{m}$ and $W_{\text{mmi}} = 2.5 \mu\text{m}$ for compact footprint and large fabrication tolerance. Figures 3(b) and 3(c) show the propagation mode field for fundamental TE polarization and TM polarization at 1550-nm wavelength. One can find that the twofold images of the two polarizations are located near the start of the output tapers. Hence, such a compact device can obtain polarization-independent 3-dB power splitter with low loss. The calculated EL s spectra are shown in Fig. 3(d). The maximum EL s of 0.6 dB for TE polarization and 0.33 dB for TM polarization occur at 1675-nm wavelength.

The proposed two types of 1×2 power splitters are fabricated on a standard SOI platform with a 220-nm core silicon layer and 2- μm buried oxide layer. The patterns are defined on a negative resist (maN-2403) using E-beam lithography (EBL, Raith 150 II). Then, inductively coupled plasma reactive ion etching (ICP-RIE) is used to transform the pattern into the 220-nm silicon layer by one etching step. The fully etched TM gratings are also fabricated in this step. Then the 70-nm etched TE gratings are fabricated by another overlay EBL exposure with PMMA resist and a 70-nm shallow etching step. Finally, a 1- μm -thick silicon oxide cladding is deposited by plasma-enhanced chemical vapor deposition (PECVD). The parameters (n , L_{mmi}) of the fabricated ARPSs are (1, 4.05 μm), (0.71, 4.12 μm), (0.53, 4.21 μm), (0.40, 4.66 μm), and (0.01, 4.81 μm) for PS s equal to 0.5, 0.6, 0.7, 0.8, and 0.9, respectively. Figure 4(a) shows the optical microscopy image of one fabricated device with $PS = 0.5$. The scanning electron microscopy (SEM) images are shown in Figs. 4(b)–4(f). To verify the robustness of the present ARPSs, we also fabricate another two groups of ARPSs on the same chip with core-width variation $\Delta W = 40 \text{ nm}$ and -40 nm , respectively. An amplified spontaneous emission (ASE) source and an optical spectrum analyzer (OSA) are used to characterize the fabricated devices. The TE-polarized light is launched to the devices through input grating I1, and the OSA is applied to record the transmissions at O1 and O2. There are straight waveguides fabricated on the same chip for normalization. Then, the PS s and EL s could be calculated according to Eqs. (1) and (2).

The measured EL s spectra are shown in Fig. 5(a). The EL s spectra fluctuate between -0.25 dB and 0.69 dB , which is attributed to the random fabrication errors of the grating and the measurement errors. At 1550 nm, the measured EL s for the five devices are between -0.08 dB and 0.58 dB , with

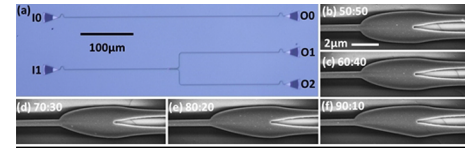


Fig. 4. (a) Optical microscopy of the fabricated ARPS with PS of 0.5 and reference waveguide; SEM images of the fabricated ARPSs with PS of (b) 0.5, (c) 0.6, (d) 0.7, (e) 0.8, and (f) 0.9.

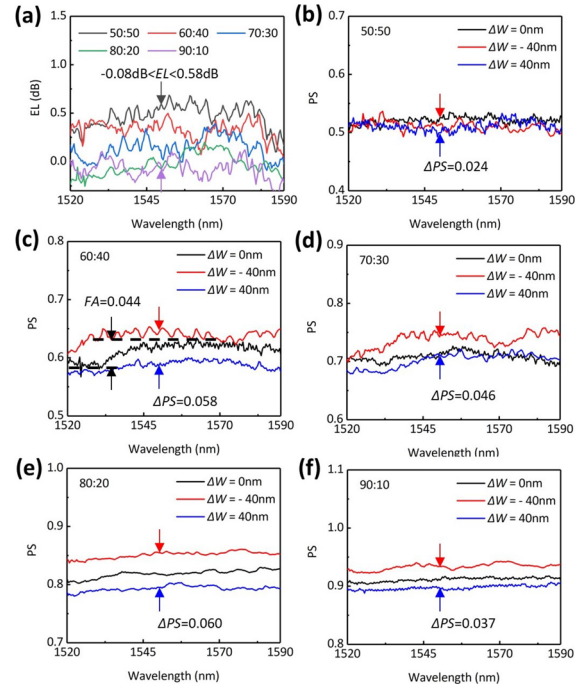


Fig. 5. (a) Measured EL s spectra for PS s equal to 0.5, 0.6, 0.7, 0.8, and 0.9; the measured PS s spectra with five power splitting ratios, including (b) 0.5, (c) 0.6, (d) 0.7, (e) 0.8, and (f) 0.9, as well as another two groups of ARPSs with core-width variation ΔW of 40 nm and -40 nm , respectively.

an average EL of 0.17 dB, which agrees well with the simulation. Figures 5(b)–5(f) show the measured PS s spectra with five power splitting ratios, including 0.5, 0.6, 0.7, 0.8, 0.9, as well as another two groups of ARPSs with core-width variation $\Delta W = 40 \text{ nm}$ and -40 nm , respectively. For $\Delta W = 0 \text{ nm}$, the maximum FA is obtained to be approximately 0.044 when $PS = 0.6$, which is marked in Fig. 5(c). The trend of PS s spectra agrees well with the simulation results shown in Fig. 2(c). For $\Delta W = 40 \text{ nm}$ and -40 nm , the FA s of PS s spectra have no significant influence compared with $\Delta W = 0 \text{ nm}$. However, the core-width variation ΔW has an influence on the symmetry of the MMI section, so the PS will change correspondingly. We mark the PS s variation at 1550 nm by the blue and red arrows in Figs. 5(b)–5(f). One can find that the ΔPS s could still be less than 0.06 even with $\Delta W = \pm 40 \text{ nm}$, which indicates a large fabrication tolerance characteristic of the proposed ARPSs.

For UDPSs, it is difficult to cover the bandwidth of 1260–1675 nm with only one grating coupler. Here, we fabricate four sets of devices close on the same chip with two kinds of TE grating couplers (TE-type1, TE-type2) and two kinds of TM

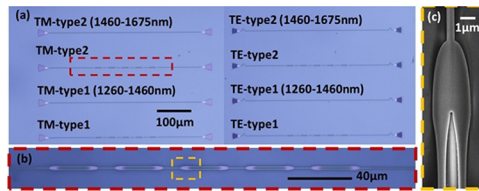


Fig. 6. (a) Optical microscopy of the fabricated UDPSSs in which the red dashed box is the position of (b) the cascaded UDPSSs; (c) SEM image of the fabricated UDPSSs.

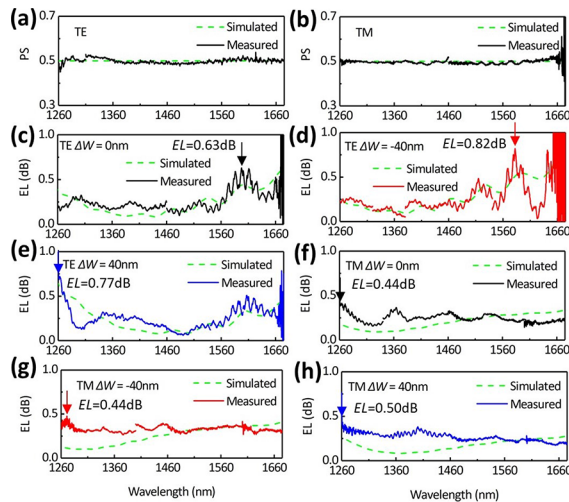


Fig. 7. Simulated and measured PS s spectra of (a) TE polarization and (b) TM polarization; simulated and measured EL s spectra of TE polarization with (c) $\Delta W = 0$ nm, (d) $\Delta W = -40$ nm, (e) $\Delta W = 40$ nm, and of TM polarization with (f) $\Delta W = 0$ nm, (g) $\Delta W = -40$ nm, (h) $\Delta W = 40$ nm.

grating couplers (TM-type1, TM-type2), as shown in Fig. 6(a). The TE-type1 and TM-type1 (TE-type2 and TM-type2) grating couplers could cover the wavelength range of 1260–1460 nm (1460–1675 nm) with incident angles of 10° , 20° , and 30° . To minimize the random errors caused by the fabrication imperfections and measurement instability, we fabricate ten identical UDPSSs cascaded in series, as shown in Fig. 6(b). Figure 6(c) shows the SEM image of the fabricated UDPSS. A supercontinuum source and an OSA are used to measure the spectra. The PS test structures are the same as Fig. 4(a). The simulated and measured PS s spectra for TE polarization and TM polarization are shown in Figs. 7(a) and 7(b). The measured PS s are within 0.46–0.53 over the whole bandwidth for both polarizations. The measured results deviate slightly from the theoretical values, which can be attributed to fabrication and measurement errors. For the convenience of comparison, the simulated and measured spectra are plotted in the same figures, as shown in Figs. 7(c)–7(h). Both the value and the trend of the measured EL s spectra are in good agreement with the simulated ones. The maximum measured EL s are marked in each figure by arrows. Within the wavelength range of 1260–1675 nm, the measured EL s could be below 0.82 dB for TE polarization and below 0.50 dB for TM polarization, even when the width variation ΔW is as large as ± 40 nm. One may find that the EL s spectra suffer from larger noise with wavelength larger than 1660 nm in Figs. 7(c)–7(e), which is caused by the degradation of coupling

efficiency for the TE2 grating couplers with the incident angle of 10° .

In summary, we propose and experimentally demonstrate 1×2 power splitters with Bezier curve contours optimized MMI, which can implement ARPSs and UDPSs as examples. For ARPSs, the measured results show that the maximum FA is < 0.044 within the wavelength range from 1520 nm to 1590 nm and the average EL s of fabricated devices is 0.17 dB at 1550 nm. With $\Delta W = \pm 40$ nm, the FA of PS s spectra have no significant increase, and the variation of PS s is below 0.06 at 1550 nm. For UDPSs, the measured EL is < 0.63 dB for TE polarization and < 0.44 dB for TM polarization within the wavelength range from 1260 nm to 1675 nm. The present UDPSs also show a large width-error tolerance of ± 40 nm. The footprints of such devices are both smaller than $10 \mu\text{m} \times 2.5 \mu\text{m}$ (without input straight waveguide). Overall, the proposed 1×2 power splitters are ultra-compact, low-loss, and fabrication friendly, which could be used for high-performance components in silicon photonic integrated circuits.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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